

LLM Evaluation

LLM Evaluation means **measuring how well a Large Language Model performs** for a specific task.

1 Why LLM Evaluation is Needed

Without evaluation you cannot know if your AI is:

-  Giving **correct answers**
-  Using the **right documents from vector DB**
-  Avoiding **hallucination**
-  Responding **consistently**
-  Performing well in **production**

Example:

Bank document assistant.

User asks:

“What is the interest rate for home loan?”

Evaluation checks:

- Did the model retrieve the **correct PDF chunk**?
- Did the model generate the **correct answer**?
- Did the model **invent information**?

2 Two Types of LLM Evaluation

1. Automatic Evaluation (Metric Based)

Measured using **scores**.

Common metrics:

Metric	Purpose
BLEU	Compare generated text vs reference
ROUGE	Summary similarity

Metric	Purpose
BERTScore	Semantic similarity
Exact Match	Perfect answer match
F1 Score	QA accuracy

Example

Reference answer:

Home loan interest rate is 8.5%

Model output:

The home loan interest rate is 8.5 percent.

Evaluation score $\approx$ **very high**

2. Human Evaluation

Humans check:

- Correctness
- Relevance
- Fluency
- Safety

Example scoring:

Metric	Score
Correctness	4/5
Relevance	5/5
Clarity	4/5

3 LLM Evaluation in RAG Systems

When you use **RAG architecture**, evaluation has **3 layers**.

1 Retrieval Evaluation

Check if the **vector database retrieved correct documents**.

Metrics:

- Precision
- Recall
- Hit Rate

Example:

Query:

Insurance claim waiting period

Vector DB retrieves:

- Policy document
- Claim rules

If correct document appears → **good retrieval**

2 Generation Evaluation

Check if **LLM answer is correct**.

Metrics:

- Faithfulness
- Relevance
- Hallucination rate

Example:

Document says:

Waiting period is 30 days

Model says:

Waiting period is 60 days

✗ Hallucination

3 End-to-End Evaluation

Check full pipeline:

User → Retrieval → LLM → Final Answer

Evaluate:

- Answer correctness
 - Context usage
 - Latency
-

Popular LLM Evaluation Tools

◆ 1. LangSmith

Used with **LangChain**

Features:

- Trace LLM calls
 - Debug RAG pipelines
 - Evaluate prompts
 - Dataset testing
-

◆ 2. RAGAS

Specifically designed for **RAG evaluation**.

Metrics:

- Faithfulness
 - Answer relevance
 - Context precision
 - Context recall
-

◆ 3. DeepEval

LLM testing framework.

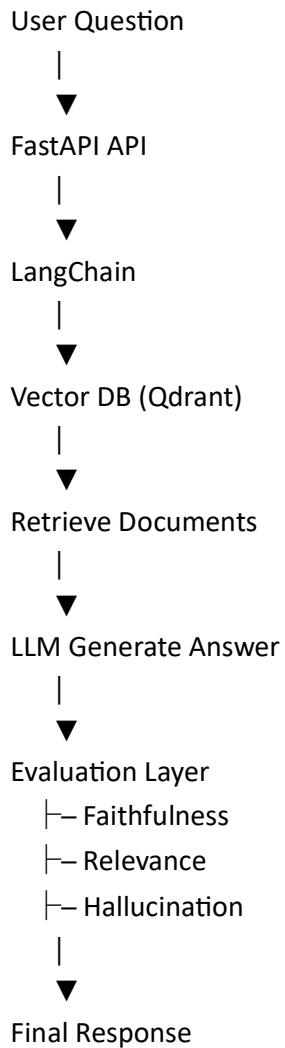
Used for:

- Unit testing LLM
- Prompt testing
- Hallucination detection

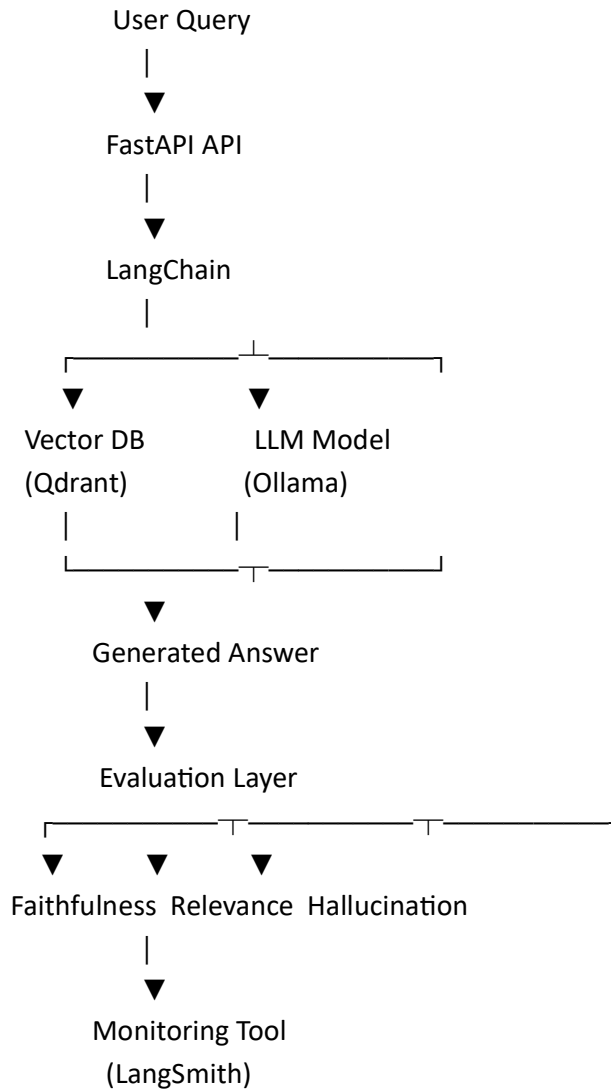
◆ 4. OpenAI Evals

Benchmark testing framework for LLM performance.

5 Example LLM Evaluation Pipeline



6 Enterprise LLM Evaluation Architecture



7 Real Enterprise Example

Bank AI Assistant

Evaluation checks:

Metric	Example
Retrieval	Did correct policy doc load
Faithfulness	Did answer match document
Latency	< 2 seconds

Metric	Example
Safety	No wrong financial advice

8 Simple Example with RAGAS

Python example:

```
from ragas.metrics import faithfulness, answer_relevancy
from ragas import evaluate
```

```
result = evaluate(
    dataset,
    metrics=[faithfulness, answer_relevancy]
)
```

```
print(result)
```

9 What Companies Actually Track

Metric	Target
Faithfulness	> 0.9
Answer relevance	> 0.85
Hallucination rate	< 5%
Response latency	< 2 sec

✓ Simple definition

LLM Evaluation is the process of measuring whether an AI model gives **accurate, relevant, and trustworthy answers**.

💡 Since you are learning **FastAPI + LangChain + Qdrant**, the **most important evaluation tool for you is:**

- **RAGAS**
- **LangSmith**